FINAL EXPLANATION: ESSENTIAL MATHEMATICS GRADE 10

Student Assessment Document

FINAL EXPLANATION: ESSENTIAL MATHEMATICS – GRADE 10 | SUB-STRAND 2.2: REFLECTION	
Student Assessment Document	
Student Name	____
Class	____
Date	____

INSTRUCTIONS FOR STUDENTS

FINAL EXPLANATION TASK — STUDENT INSTRUCTIONS

Driving Question: Why does a mirror show a perfect copy of the world — and where exactly does that copy live?

You have spent 8 lessons investigating the geometric properties of reflection using plane mirrors — from Amina's bedroom mirror and the matatu's rear-view glass to the letter "A" your teacher held up in class. Now it is time to pull everything you have learned together into one well-reasoned, evidence-based explanation.

Your Final Explanation should answer the driving question fully and precisely. Think of it as your chance to teach someone else — perhaps a younger student at your school — exactly why a mirror produces what looks like a "perfect copy," and exactly where that copy lives.

HOW TO USE THIS DOCUMENT:

- Work through each section in order. Each section has a guiding prompt and a model exemplar answer to show you the standard expected.
- Write your own response to each prompt in your exercise book or on the answer sheet provided before reading the exemplar.
- Use correct mathematical vocabulary: mirror line / axis of reflection, object, image, perpendicular, equidistant, congruent, lateral inversion.
- Support every claim with geometric reasoning — do not rely on phrases like "it just looks that way."
- Where instructed, include a clearly labelled diagram.

REMEMBER: A strong Final Explanation does three things — it states a precise geometric rule, it shows why that rule is true using evidence or reasoning, and it connects back to a real example from the phenomenon (Amina's mirror, the matatu mirrors, or the letter "A").

Section 1 — The Mirror Line: The Invisible Rule That Governs Reflection	
Explain what a mirror line (axis of reflection) is and the role it plays when an object is	When Amina stands in front of her bedroom mirror, the surface of the glass acts as the mirror line — the fixed line of reflection. Every point on Amina (the object) is mapped to a matching point in her reflection (the image) according to two strict distance rules.

reflected. In your explanation, answer: What is the mirror line? What two distance properties does it enforce? Use Amina's bedroom mirror as your context, and include a labelled diagram showing a point P, its image P′, the mirror line, and the perpendicular connector between them.	Rule 1 — Equal perpendicular distances: The distance from any object point P to the mirror line, measured along the perpendicular to that line, is exactly equal to the distance from the mirror line to the corresponding image point P′. In notation: if M is the foot of the perpendicular from P to the mirror line, then PM = MP′. Rule 2 — Perpendicularity: The line segment PP′ always meets the mirror line at a right angle (90°). Diagram (draw and label yours): — Draw a vertical line labelled 'Mirror line (m)'. — Mark a point P to the left of the line and its image P′ the same distance to the right. — Draw the horizontal segment PP′, marking the right-angle symbol where it crosses m. — Label PM and MP′ as equal lengths. This is why Amina notices that she appears to stand as far behind the mirror as she actually stands in front of it: her image is always equidistant from the mirror line on the opposite side.

Section 2 — Properties of a Reflected Image: Size, Shape, and Orientation	
List and explain all four key properties of the image formed by reflection in a plane mirror. For each property, state whether the property is preserved or changed, give the correct mathematical term, and link it to at least one observation from the phenomenon — Amina's uniform, the matatu mirrors, or the letter 'A'.	When an object is reflected in a plane mirror, the image has the following four properties: 1. SAME SIZE (Lengths are preserved — congruence): Every length in the image equals the corresponding length in the object. This is why the pattern on Amina's Kanga skirt and the number of buttons on her shirt appear identical in the mirror — no stretching or shrinking occurs. Mathematically, reflection is an isometry (a distance-preserving transformation). 2. SAME SHAPE (Angles are preserved): Every angle in the image equals the corresponding angle in the object. The triangles, curves, and corners of Amina's reflection are geometrically congruent to those on her actual uniform. 3. LATERALLY INVERTED (Orientation is reversed): Although size and shape are preserved, the image is flipped — left and right are swapped across the mirror line. This is called lateral inversion. It explains why Amina raises her right hand but her reflection appears to raise its left hand, and why the letter 'A' placed beside a mirror appears as a horizontally flipped 'A'. Note: the letter 'A' is symmetrical, so the flip is less obvious — try it with the letter 'R' and the inversion becomes very clear. 4. IMAGE IS VIRTUAL (The image appears behind the mirror): The reflected image cannot be projected onto a screen because light rays do not actually converge behind the mirror — they only appear to. This is why the copy 'lives' behind the glass: it is a virtual image, located as far behind the mirror line as the object is in front.

<table>
<tr><td></td><td>Summary table (copy into your notes): | Property	Preserved or Changed?	Term used
Lengths	Preserved	Congruence / Isometry
Angles	Preserved	Congruence
Orientation	Changed	Lateral inversion
Position	Image behind mirror	Virtual image |</td></tr>
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<tr><th colspan="2">Section 3 — Constructing the Image: The Step-by-Step Geometric Method</th></tr>
<tr><td>Describe, step by step, how you would accurately construct the reflection of a triangle ABC in a given mirror line m on a square-grid or plain paper. Your explanation should be precise enough for another Grade 10 student to follow without any additional help. State which tools you need and why each step guarantees the image is correct.</td><td>Tools needed: ruler, set square (or protractor), pencil, eraser.

Step 1 — Identify the object points.
Label the three vertices of your triangle A, B, and C clearly on your paper.

Step 2 — Draw perpendiculars from each vertex to the mirror line m.
From vertex A, use your set square to draw a line that meets m at exactly 90°. Mark the foot of this perpendicular as A_m. Repeat for vertices B and C to get feet B_m and C_m. (The right angle guarantees we are measuring the shortest — perpendicular — distance, which is what the rule requires.)

Step 3 — Mark the image points on the opposite side.
Measure the distance AA_m with your ruler. Mark point A′ on the other side of m along the same perpendicular line so that $A_mA' = AA_m$. Repeat to find B′ and C′.

Step 4 — Connect the image points.
Join A′, B′, and C′ to form triangle A′B′C′. This is the reflected image.

Step 5 — Verify.
Check that: (a) each image vertex is the same distance from m as its object vertex; (b) the line joining each object–image pair (e.g. AA′) is perpendicular to m; (c) the image triangle is congruent to the original (measure corresponding sides and angles).

Why this works: Every step mirrors the two fundamental rules — equal perpendicular distances and perpendicularity of the connector — that govern reflection. These rules ensure the image is an exact congruent copy, laterally inverted, on the opposite side of m.</td></tr>
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<tr><th colspan="2">Section 4 — Lines of Symmetry: When the Mirror Line Lives Inside the Shape</th></tr>
<tr><td>Explain what a line of symmetry is and how it connects to the</td><td>A line of symmetry is a mirror line placed through a shape such that the reflection of the shape in that line maps the shape exactly onto itself — every point on one side has a matching point on the other side at an equal perpendicular distance. In other words, a</td></tr>
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geometric definition of reflection. Identify and justify all lines of symmetry for: (a) an equilateral triangle, (b) a rectangle, and (c) the letter 'A' as seen in the classroom phenomenon. Explain why a scalene triangle has no line of symmetry.	line of symmetry is a mirror line for which the image and the object are identical and perfectly overlap. (a) Equilateral triangle — 3 lines of symmetry. Each line of symmetry runs from one vertex to the midpoint of the opposite side. Because all three sides are equal and all three angles are 60°, folding along any of these three lines produces two halves that match perfectly. (b) Rectangle — 2 lines of symmetry. A rectangle has one line of symmetry joining the midpoints of the longer pair of opposite sides, and one joining the midpoints of the shorter pair. Note: the diagonals of a rectangle are NOT lines of symmetry (folding along a diagonal does not make the two halves coincide, because the sides are unequal). (c) The letter 'A' — 1 line of symmetry. The vertical line running through the apex and the midpoint of the crossbar divides the 'A' into two mirror-image halves. This is exactly what the class observed: when the teacher placed a mirror along this vertical axis, the reflection completed the letter perfectly. Why a scalene triangle has no line of symmetry: In a scalene triangle, all three sides have different lengths and all three angles are different. Any line you draw through the triangle will produce two halves that cannot be reflected onto each other — the unequal sides mean the equal-distance rule is violated for at least some points. Therefore, no line of symmetry exists.

Section 5 — Answering the Driving Question: The Full Explanation	
Write a complete, connected answer to the driving question: 'Why does a mirror show a perfect copy of the world — and where exactly does that copy live?' Your answer must: (1) state the geometric rule that governs reflection; (2) explain why the copy is the same size and shape but laterally inverted; (3) state precisely where the image is located and why it appears to be behind the mirror; (4) reference at least TWO specific observations from the phenomenon (Amina, the matatu, or the letter 'A'). Aim for a well-organised paragraph response of 150–200 words.	A plane mirror shows a perfect copy of the world because of a precise geometric rule: every point on an object is mapped to a corresponding image point on the opposite side of the mirror line, at an equal perpendicular distance from that line. This rule guarantees that all lengths and angles are preserved — making the image congruent to the object — which is why Amina's Kanga skirt pattern and the number of buttons on her shirt appear identical in her bedroom mirror. However, because the image is flipped across the mirror line, left and right are swapped — this lateral inversion explains why Amina raises her right hand but her reflection appears to raise its left, and why the letter 'A' on the teacher's paper appears reversed when held beside a mirror. The copy 'lives' behind the mirror because light rays reflecting off the mirror surface appear to the eye as though they are diverging from a point directly behind the glass, as far behind the mirror as the object is in front. This is a virtual image — it cannot be caught on a screen, but it obeys the same equal-distance rule perfectly. The same principle explains the images seen in the matatu's rear-view mirror: every vehicle behind is shown as a clear, equidistant virtual image, laterally inverted across the mirror's surface.

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Criterion	**Excellent (4)**	**Proficient (3)**	**Developing (1–2)**
Geometric Accuracy — Mirror Line Properties	Clearly and correctly states both properties of reflection (equal perpendicular distances; PP′ perpendicular to mirror line) with precise mathematical language. Diagram is fully and correctly labelled.	States both properties correctly with mostly accurate language. Diagram is present and mostly correct, with minor labelling omissions.	States only one property or describes properties in vague, non-geometric terms (e.g. 'the image is the same distance away'). Diagram is missing or significantly incorrect.
Properties of the Reflected Image	Accurately identifies and explains all four properties (congruence of lengths, congruence of angles, lateral inversion, virtual image) using correct mathematical terminology. Each property is linked to a specific observation from the phenomenon.	Identifies and explains at least three of the four properties correctly. Links at least two properties to the phenomenon. Minor errors in terminology.	Identifies fewer than three properties, or descriptions are largely informal and lack mathematical terms. Little or no connection to the phenomenon.
Construction Method	Provides a clear, complete, step-by-step construction method that a peer could follow independently. Correctly identifies all tools and explicitly justifies why each step ensures geometric accuracy.	Provides a mostly complete construction method with steps in logical order. Justification for at least two steps is given. Minor gaps or ambiguities present.	Steps are incomplete, out of order, or lack geometric justification. Another student could not reliably reproduce the construction from the description given.
Lines of Symmetry	Correctly identifies and justifies the number of lines of symmetry for all three given shapes and provides a valid geometric reason why a scalene triangle has none. Justifications reference the equal-distance reflection rule.	Correctly identifies lines of symmetry for at least two of the three shapes. Gives a partially correct reason for the scalene triangle. Justifications are present but may not explicitly reference the reflection rule.	Correctly identifies lines of symmetry for at most one shape, or counts incorrectly (e.g. includes diagonals as lines of symmetry for a rectangle). Little or no justification provided.
Driving Question Response — Coherence and Completeness	Provides a fully coherent, well-organised response (150–200 words) that addresses all four required elements: the governing rule, size/shape preservation with lateral inversion, precise location of the virtual image, and at least two specific phenomenon references. Mathematical vocabulary is used accurately throughout.	Addresses at least three of the four required elements in a mostly coherent response. At least one phenomenon reference is included. Some mathematical vocabulary is used correctly.	Addresses fewer than three required elements. Response is largely informal or narrative, with little mathematical vocabulary. Phenomenon connections are vague or absent.